

Supplementary Material for Penalized Fair Regression for Multiple Groups in Chronic Kidney Disease by Carter Nakamoto, Lucia Chen, Agata Foryciarz, and Sherri Rose

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Short Running Head: Penalized Fair Regression for Multiple Groups

Abstract

Web Appendix A: Continuous Outcomes

Here we seek to estimate continuous outcome Y . The baseline continuous outcome regression generates estimates $\hat{Y}$ as a function of m input variables X , indexed by q , by identifying the estimator that minimizes the mean squared error (MSE) loss. The unfairness metric is mean residual disparity, defined as the difference in mean residuals between the reference group and the group of interest:

$$U_{MR} = \frac{1}{n_r} \sum_{k \in r} (\hat{Y}_k - Y_k) - \frac{1}{n_g} \sum_{j \in g} (\hat{Y}_j - Y_j). \quad (1)$$

Penalized regression with continuous outcomes will add penalties for unfairness as defined in Equation 1, where each group has its own penalty term and penalty weight.

Example: Lasso

We first use the example of lasso, which involves adding a penalty on regression coefficients β with penalty weight γ to the mean squared error loss and estimating $\hat{Y}$ as a linear combination of regression coefficients and predictors:

$$L = \sum_{i=1}^n \left[\sum_{q=1}^m \beta_q X_{iq} - Y_i \right]^2 + \gamma \sum_{q=1}^m |\beta_q|.$$

The loss function with fairness penalties is given by:

$$L = \sum_{i=1}^n \left[\sum_{q=1}^m \beta_q X_{iq} - Y_i \right]^2 + \gamma \sum_{q=1}^m |\beta_q| + \sum_{g \in G} \lambda_g \left[\frac{1}{n_r} \sum_{k \in r} (\hat{Y}_k - Y_k) - \frac{1}{n_g} \sum_{j \in g} (\hat{Y}_j - Y_j) \right].$$

This loss function minimization problem can be solved using the L-BFGS algorithm, a type of quasi-Newtonian optimizer (Liu and Nocedal, 1989), available in the PyTorch package.

Example: Neural Net

A penalized loss function for a simple neural network with two hidden layers could be defined as:

$$L = \sum_{i=1}^n \left[\sum_{q=1}^m \beta_q X_{iq} - Y_i \right]^2 + \sum_{g \in G} \lambda_g \left[\frac{1}{n_r} \sum_{k \in r} (\hat{Y}_k - Y_k) - \frac{1}{n_g} \sum_{j \in g} (\hat{Y}_j - Y_j) \right]. \quad (2)$$

The loss function minimization problem here can be solved using the Adam algorithm, a type of gradient descent-based optimizer ([Loshchilov and Hutter, 2019](#)), also available in the PyTorch package.

Web Appendix B: Weighted Classification Derivation

We begin with the loss function from section 3.1 of the manuscript.

$$L = - \sum_{i=1}^n \{Y_i \ln(\hat{P}_i) + (1 - Y_i) \ln(1 - \hat{P}_i)\} + \sum_{g \in G} \lambda_g n_g \left(\frac{\sum_{k \in r} Y_k \hat{P}_k}{\sum_{k \in r} Y_k} - \frac{\sum_{j \in g} Y_j \hat{P}_j}{\sum_{j \in g} Y_j} \right).$$

We could calculate the gradient and use a gradient descent-based algorithm to find the solution. With $\nabla \hat{P} = \hat{P}(1 - \hat{P})X$, the gradient is

$$\nabla L = - \sum_{i=1}^n (Y_i - \hat{P}_i) X_i + \sum_{g \in G} \lambda_g n_g \left(\frac{\sum_{k \in r} \nabla \hat{P}_k}{\sum_{k \in r} Y_k} - \frac{\sum_{j \in g} Y_j \nabla \hat{P}_j}{\sum_{j \in g} Y_j} \right).$$

However, there is a more efficient method available based on the reductions approach of [Agarwal et al. \(2018\)](#). In this approach, we use cost minimization loss, which punishes misalignment between $\hat{P}_i$ and Y_i with a cost and minimizes total costs ([Zadrozny et al., 2003](#)). The cost-sensitive loss function with the standard erroneous classification costs of 1 for all observations, which has the same solution as the standard binary cross entropy loss, can be added to the true positive rate disparity unfairness penalty terms to get the cost

minimization penalized loss function:

$$L = \sum_{i=1}^n \left\{ \hat{P}_i(1 - Y_i) + (1 - \hat{P}_i)Y_i \right\} + \sum_{g \in G} \lambda_g n_g \left(\frac{\sum_{k \in r} Y_k \hat{P}_k}{\sum_{k \in r} Y_k} - \frac{\sum_{j \in g} Y_j \hat{P}_j}{\sum_{j \in g} Y_j} \right).$$

This penalized loss function is, itself, a cost-sensitive classification loss function with more complex costs. This is apparent when the function is rewritten:

$$L = \sum_{i=1}^n \left[\hat{P}_i \left\{ 1 - Y_i + Y_i \sum_{g \in G} \lambda_g n_g \left(\frac{\mathbb{I}_{ri}}{\sum_{k \in r} Y_k} - \frac{\mathbb{I}_{gi}}{\sum_{j \in g} Y_j} \right) \right\} + (1 - \hat{P}_i)Y_i \right].$$

Reconceptualizing of the penalized fair regression as a cost-sensitive classification problem, the costs for positive (C_i^1) and negative (C_i^0) classifications are

$$C_i^1 = 1 - Y_i + Y_i \sum_{g \in G} \lambda_g n_g \left(\frac{\mathbb{I}_{ri}}{\sum_{k \in r} Y_k} - \frac{\mathbb{I}_{gi}}{\sum_{j \in g} Y_j} \right),$$

and $C_i^0 = Y_i$. These costs differ from the standard cost of 1 for all incorrect classifications and 0 for all correct classifications. For reference group individuals, even correct positive classifications incur some cost, and for individuals in the groups $g \in G$, correct positive classifications will incur negative costs, which could also be called rewards.

With this formulation, we extend a theorem from [Zadrozny et al. \(2003\)](#) to rewrite the cost-sensitive classification problem as a weighted non-cost-sensitive classification problem.

In this formulation, the weights are

$$W_i = |C_i^0 - C_i^1| = \left| Y_i \left\{ 2 - \sum_{g \in G} \lambda_g n_g \left(\frac{\mathbb{I}_{ri}}{\sum_{k \in r} Y_k} - \frac{\mathbb{I}_{gi}}{\sum_{j \in g} Y_j} \right) \right\} - 1 \right|,$$

and modified outcomes are $Y'_i = \mathbb{I}(C_i^0 > C_i^1)$. It is then possible to perform the desired fair penalized regression with a wide array of standard classifiers (e.g., logistic regression, support vector machines, decision trees) that are compatible with individual-level sample weighting.

Web Appendix C: Simulation Study

Let g_A , g_B , and g_C be the three different groups whose membership was reflected by indicators $\mathbb{I}_{g_A}$, $\mathbb{I}_{g_B}$, and $\mathbb{I}_{g_C}$, respectively. We defined the reference group to be the set of individuals who were not in any of the three groups. The data were generated as follows with $n = 1,000,000$:

$$X_1, X_2 \sim \mathcal{N}(30, 15)$$

$$X_3, X_4 \sim \text{Pois}(15)$$

$$X_5, \dots, X_9 \sim \text{Bern}(p), p = [0.2, 0.4, 0.6, 0.8, 0.95]$$

$$\mathbb{I}_{g_A} \sim \text{Bern}(\min[\max\{c_0(X_9 + 0.1) + c_3X_6 + \mathcal{N}(0, 0.02), 0\}, 1])$$

$$\mathbb{I}_{g_B} \sim \text{Bern}(\min[\max\{c_1(X_7 + 0.05) + c_4X_6 + \mathcal{N}(0, 0.02), 0\}, 1])$$

$$\mathbb{I}_{g_C} \sim \text{Bern}(\min[\max\{c_2(X_8 + 0.4) + c_5X_6 + \mathcal{N}(0, 0.02), 0\}, 1])$$

$$Y \sim \text{Bern}\left(\Phi^{-1}\left[\frac{X_1}{30} - \frac{X_2}{15} + \frac{3X_3}{50} - \frac{X_4}{25} + \frac{1}{100}\left\{2^{X_5+X_6} + 6X_7 + 10X_8X_9 + 8X_3A + 6X_4B + C\mathcal{N}(60, 2)\right\} + \mathcal{N}(0, 1)\right]\right).$$

Across three simulation settings, $c = c_0, \dots, c_5$ vary. Simulation setting 1 has all moderately sized groups with $c = (0.2, 0.2, 0.2, 0.02, 0.02, 0.02)$, g_A is small in simulation setting 2 with $c = (0.02, 0.2, 0.2, 0.002, 0.02, 0.02)$, and all groups are small in simulation setting 3 with $c = (0.02, 0.02, 0.02, 0.002, 0.002, 0.002)$. There is substantial overlap between the groups g_A , g_B , and g_C in all settings.

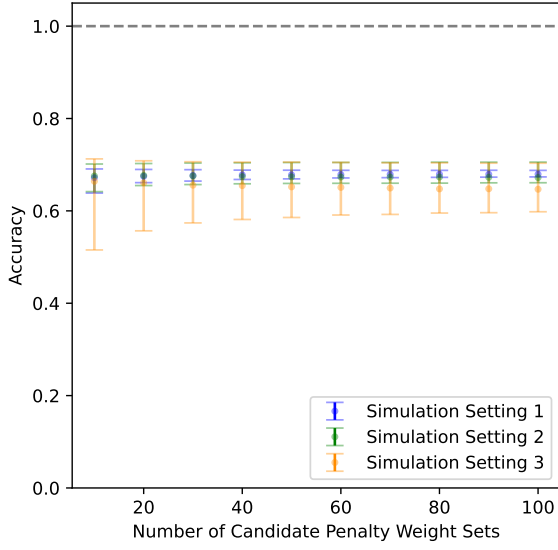
Web Table 1: **Identification Strategy for ABFM Variables.** For most variables, we queried for the enumerated OMOP concepts in the enumerated fields. For other variables, we queried the OMOP concept lookup table for the OMOP ABFM data with the given text.

Diagnosis	Search Method	Query
ESRD	Text Query	“End stage renal disease” or “End stage renal failure”
eGFR	Preexisting	See repository
HbA1c	OMOP Concepts: Measurement	4184637, 36304734
Bilirubin	OMOP Concepts: Measurement	4118986, 3024148, 4216632, 3024128
Diastolic BP	OMOP Concepts: Measurement	4298393, 3012888
Systolic BP	OMOP Concepts: Measurement	4152194, 3004249
CKD	OMOP Concepts: Condition	44784621, 443611, 443601, 44782690, 43021852, 45768812, 44782728, 45763855, 45763854, 43531578, 45768812, 443612, 44782429, 43531653, 443614, 43021852, 443597
Diabetes (Type 2)	OMOP Concepts: Condition / ICD10	44809548, 37208172 / E11

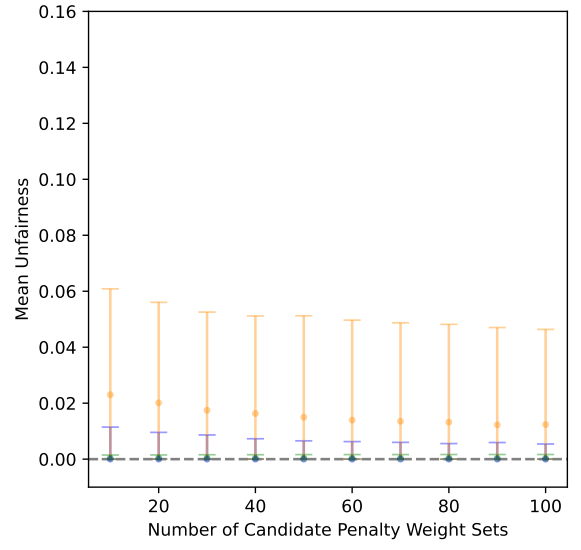
Diagnosis	Search Method	Query
Kidney Stone	OMOP Concepts: Condition	201620, 45548655, 44822046
Obstruction	OMOP Concepts: Condition	45612141, 45592125, 45567838, 36713449, 4220439, 433813
Hypertension	OMOP Concepts: Condition	43020424, 45768449, 37200492, 37200493, 37200494
Proteinuria	Text Query	“Proteinuria”
Glomerulonephritis	Text Query	“Glomerulonephritis”
Pyelonephritis	Text Query	“Pyelonephritis”
Female	OMOP Concepts: Gender	8532
White	OMOP Concepts: Race	8527, 38003614
Black	OMOP Concepts: Race	8516, 38003597, 38003598, 38003599, 38003600
Asian	OMOP Concepts: Race	8515, 38003574-38003596
Hispanic	OMOP Concepts: Ethnicity	38003563

Web Table 2: **Cohort Patient Characteristics in 2017-2018 by ESRD Status in 2019.**

	Missing	Overall	No ESRD	ESRD
n	0	67397	64593	2804
Measurements				
Minimum eGFR, mean (SD)	36899	48.3 (18.7)	49.5 (18.0)	22.3 (16.1)
Mean Diastolic BP, mean (SD)	0	75.3 (7.9)	75.4 (7.9)	74.2 (8.8)
Mean Systolic BP, mean (SD)	0	136.6 (11.9)	136.5 (11.7)	140.0 (14.1)
Mean HbA1c, mean (SD)	44250	6.8 (1.4)	6.8 (1.4)	6.9 (1.6)
Mean Bilirubin, mean (SD)	34493	0.6 (0.6)	0.6 (0.6)	0.5 (0.3)
Diagnoses				
Diabetes (Type 2), n (%)	0	47447 (70.4)	45780 (70.9)	1667 (59.5)
Kidney Stone, n (%)	0	32734 (48.6)	30947 (47.9)	1787 (63.7)
Proteinuria, n (%)	0	1255 (1.9)	1223 (1.9)	32 (1.1)
Pyelonephritis, n (%)	0	494 (0.7)	451 (0.7)	43 (1.5)
Demographics				
Age, mean (SD)	0	75.0 (11.3)	75.1 (11.3)	72.0 (12.3)
Female, n (%)	0	35655 (52.9)	34283 (53.1)	1372 (48.9)
White, n (%)	0	53402 (79.2)	51519 (79.8)	1883 (67.2)
Black, n (%)	0	8094 (12.0)	7558 (11.7)	536 (19.1)
Asian, n (%)	0	1427 (2.1)	1325 (2.1)	102 (3.6)
Hispanic non-white, n (%)	0	2678 (4.0)	2476 (3.8)	202 (7.2)

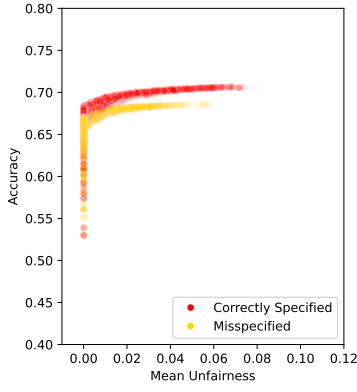


(a) Accuracy

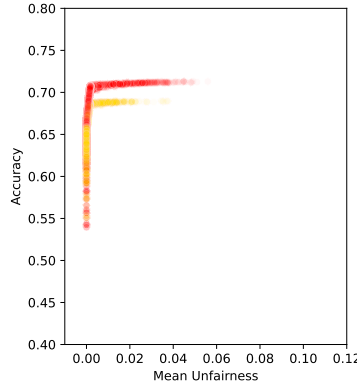


(b) Population-Weighted Unfairness

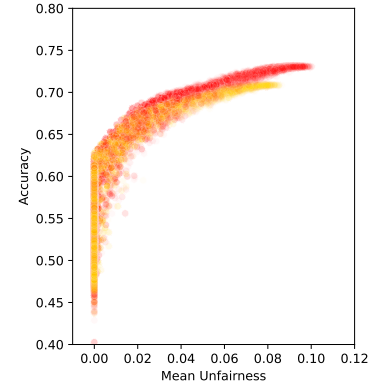
Web Figure 1: **Fair Regression Estimator Accuracy and Unfairness by Number of Candidate Unfairness Penalty Weight Draws Across Different Simulation Settings.** Confidence intervals represent the empirical 25th and 75th percentiles of replications.



(a) Simulation Setting 1



(b) Simulation Setting 2



(c) Simulation Setting 3

Web Figure 2: **Fair Regression Estimator Accuracy and Fairness for Correctly Specified and Misspecified Fair Regression Estimators by Simulation Setting.**

Each plotted point corresponds to the result of a single simulation experiment under one of the score function variations.

References

- Agarwal, A., Beygelzimer, A., Dudík, M., Langford, J. and Wallach, H. (2018). A Reductions Approach to Fair Classification. arXiv:1803.02453 [cs].
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- Zadrozny, B., Langford, J. and Abe, N. (2003). Cost-sensitive learning by cost-proportionate example weighting. In *Third IEEE International Conference on Data Mining*, pages 435–442.